context-mode — Full Technical Documentation

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> Ported from mksglu/context-mode by @mksglu (Elastic License 2.0). Core algorithms, database schemas, search pipeline, sandbox executor architecture, session event system, and compaction snapshot builder are derived from that project and adapted for the Cowork plugin architecture.

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1. Overview

context-mode is a Cowork plugin for Claude Code that reduces context window consumption by up to 98%. It does this through five mechanisms:

- **Automatic Tool Routing**: PreToolUse hooks intercept Bash, Read, Grep, WebFetch, Agent, and Task calls before they execute and redirect them through the context-mode sandbox — without changing how Claude works.
- **Sandbox Execution**: Runs code in isolated subprocesses, returning only stdout. Raw file contents and command output never enter context. Supports 11 languages.
- **Knowledge Base**: Indexes content into a local SQLite FTS5 database. Retrieves only relevant snippets via BM25 + trigram dual-strategy search.
- **Session Continuity**: Captures session events via hooks, builds priority-tiered snapshots before compaction, and restores session state afterward. A bootstrapper with dependency self-healing ensures the server starts cleanly on every session.
- **Main Skill**: The `context-mode` skill provides an in-session decision tree, tool-selection patterns, and anti-patterns so Claude consistently picks the right tool.

Problem Statement

Long Claude Code sessions in Cowork consume context rapidly. Every file read, web fetch, and shell command dumps raw output into the context window. A single Playwright snapshot can cost 56 KB. Twenty fetched documents can cost hundreds of KB. Over a long task, this bloats context, reduces response quality, and forces premature compaction — causing Claude to lose track of progress.

Solution

context-mode intercepts data-heavy operations and processes them outside the context window:

```
Without context-mode:                 With context-mode:

┌──────────────────────┐              ┌──────────────────────┐
│ Context Window       │              │ Context Window       │
│                      │              │                      │
│ [60KB file content]  │              │ [120B search result] │
│ [45KB web page]      │              │ [80B cache hint]     │
│ [30KB cmd output]    │              │ [200B summary]       │
│                      │              │                      │
│ Total: 135KB         │              │ Total: 400B (99.7%)  │
└──────────────────────┘              └──────────────────────┘
```

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2. Install

One command:

```
npx github:scottconverse/context-mode
```

This clones the plugin to the Cowork plugin cache, registers it, installs native dependencies, and verifies that all 9 MCP tools respond. After installation, start a new session and run `/context-mode:ctx-doctor` to confirm.

Via Cowork marketplace:

```
/plugin marketplace add scottconverse/context-mode
/plugin install context-mode@scottconverse-context-mode
```

Manual install:

```
git clone https://github.com/scottconverse/context-mode.git
cd context-mode
node install.js
```

Requirements:

- Node.js >= 18
- Claude Code in Cowork
- Git Bash (Windows only, for shell execution)

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3. Architecture

System Architecture

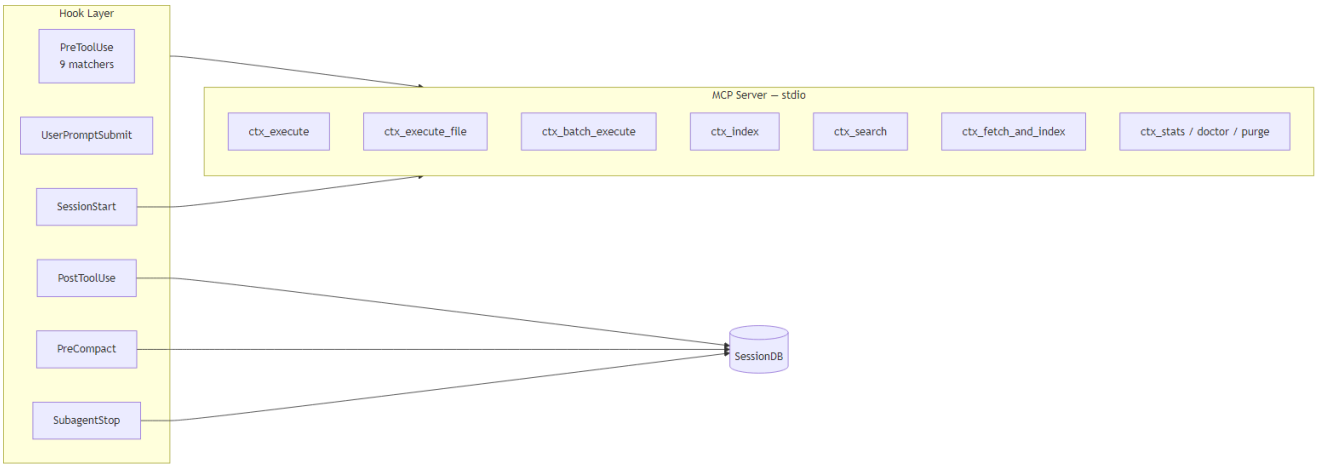

Figure 1: System Architecture — hooks, MCP server, and session database

Data Flow

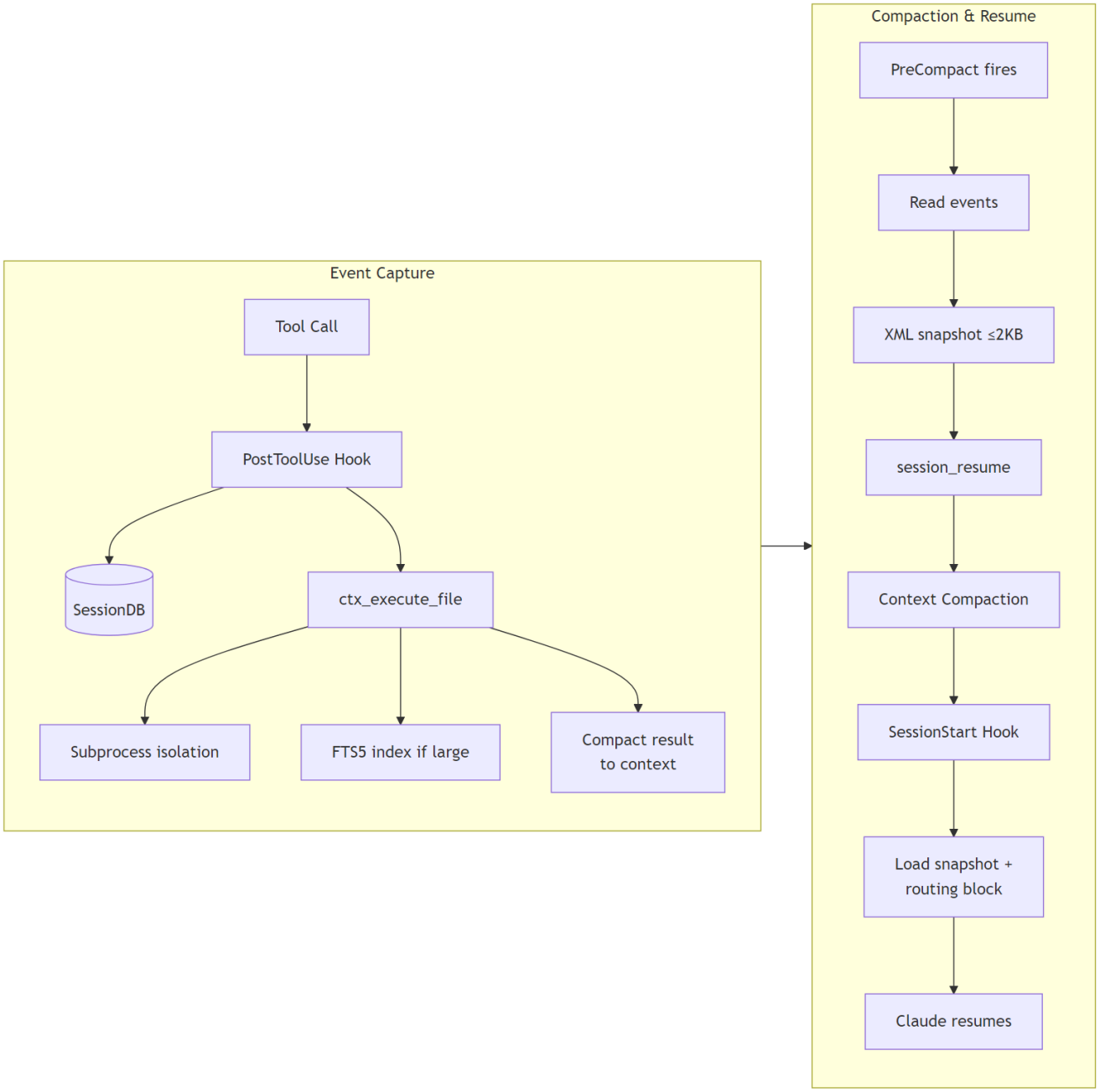

Figure 2: Data Flow — tool call through event capture, compaction, and session resume

Directory Structure

```
context-mode/
├── .claude-plugin/
│   └── plugin.json           # Cowork plugin manifest (mcpServers, hooks, skills)
├── .claude/
│   └── settings.json         # Permission rules (allow/deny)
├── .mcp.json                 # MCP server registration (flat format)
├── .gitignore
├── package.json
├── start.js                  # Bootstrapper: dep self-healing + ABI cache
├── install.js                # One-command installer with server probe
├── CLAUDE.md                 # System directive for Claude sessions
├── server/
│   ├── index.js              # MCP server (9 tools, lifecycle)
│   ├── sandbox.js            # PolyglotExecutor
│   ├── knowledge.js          # ContentStore (FTS5)
│   ├── session.js            # SessionDB
│   ├── snapshot.js           # Compaction snapshot builder
│   ├── runtime.js            # Language runtime detection
│   ├── db-base.js            # SQLite utilities
│   ├── utils.js              # Query sanitization, Levenshtein
│   └── exit-classify.js      # Non-zero exit classification
├── hooks/
│   ├── hooks.json            # 6 hook events, 14 matchers (9 PreToolUse)
│   ├── run-hook.cmd          # Cross-platform hook wrapper (CMD/bash polyglot)
│   ├── pretooluse.js         # PreToolUse: routes Bash, Read, Grep, WebFetch, Agent, Task
│   ├── posttooluse.js        # PostToolUse: captures session events (<20ms)
│   ├── precompact.js         # PreCompact: builds resume snapshot
│   ├── sessionstart.js       # SessionStart: injects routing block + session guide
│   ├── userpromptsubmit.js   # UserPromptSubmit: injects routing block per turn
│   ├── subagent-stop.js      # SubagentStop: captures sub-agent outcomes
│   ├── ensure-deps.js        # Native dependency management (ABI compatibility)
│   ├── suppress-stderr.js    # Suppresses Node.js deprecation noise
│   ├── routing-block.js      # Context routing XML template
│   ├── session-extract.js    # Event extraction (13 categories)
│   ├── session-helpers.js    # Session ID, paths, stdin
│   ├── session-directive.js  # Session guide builder
│   ├── session-loaders.js    # Shared SQLite loader factories
│   └── core/
│       ├── routing.js        # Pure routing logic (all PreToolUse decisions)
│       ├── formatters.js     # Decision → Cowork response format
│       ├── stdin.js          # Async stdin reader
│       └── tool-naming.js    # Cowork tool name generator
├── skills/
│   ├── context-mode/         # Main context-mode skill (decision tree)
│   │   ├── SKILL.md
│   │   └── references/       # anti-patterns.md, patterns-*.md
│   ├── ctx-stats/SKILL.md
│   ├── ctx-doctor/SKILL.md
│   └── ctx-purge/SKILL.md
├── agents/                   # context-optimizer agent
├── scripts/                  # setup.js, setup.sh
├── docs/                     # Landing page, full docs
└── test-e2e.js               # 207-test E2E suite (18 sections)
```

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4. Automatic Tool Routing

Introduced in v1.1.0. PreToolUse hooks intercept five built-in Claude Code tools before they execute and redirect them through the context-mode sandbox. Routing is silent and automatic — neither Claude nor the user needs to change anything.

Intercepted Tools

Intercepted Tool	Routing Decision	Redirected To
`Bash` (curl, wget, non-whitelisted)	`modify`	`ctx_execute` (sandbox)
`Bash` (git, mkdir, rm, mv, cd, ls, echo, ...)	`null` (passthrough)	Native Bash — unchanged
`Read`	`context` (one-time guidance)	Nudges toward `ctx_execute_file`; allows Read for files being edited
`Grep`	`context` (one-time guidance)	Nudges toward `ctx_execute` for sandbox search
`WebFetch`	`deny`	Redirects with guidance to `ctx_fetch_and_index`
`Agent` / `Task`	`modify`	Injects routing block into prompt

9 PreToolUse Matchers

The hooks.json registers 9 matchers for the PreToolUse event:

1. `Bash` — intercepts all Bash calls; routing logic decides allow vs. modify
2. `WebFetch` — denied; guidance directs Claude to `ctx_fetch_and_index`
3. `Read` — redirected to `ctx_execute_file`
4. `Grep` — redirected to `ctx_execute` with a grep command
5. `Agent` — routing block injected into agent prompt
6. `Task` — routing block injected into task prompt
7. `mcp__plugin_context-mode_context-mode__ctx_execute` — adds intent context
8. `mcp__plugin_context-mode_context-mode__ctx_execute_file` — adds intent context
9. `mcp__plugin_context-mode_context-mode__ctx_batch_execute` — adds intent context

Bash Whitelist

The following Bash calls are always passed through unchanged because they write state, navigate, or produce predictably small output:

```
git add / commit / push / checkout / branch / merge / tag / stash / rebase
mkdir  mv  cp  rm  touch  chmod
cd  pwd  which
kill  pkill
npm install / publish
pip install
echo  printf
```

Bash commands with curl, wget, inline HTTP, or build tools (gradle, maven) are forcibly redirected to `ctx_execute`. All other non-whitelisted Bash commands receive a one-time guidance nudge suggesting sandbox use, but are allowed to proceed.

Guidance Throttling

The first time a non-whitelisted tool is intercepted per session, Claude receives an `additionalContext` response explaining the routing decision. Subsequent intercepts of the same type are silent. This prevents guidance noise in long sessions. The throttle uses file-based atomic locking for cross-process safety (multiple Claude Code subprocesses per session).

UserPromptSubmit Hook

On every user prompt turn, the `UserPromptSubmit` hook fires and injects the current routing block into Claude's context. This ensures the model always has current tool-selection guidance, even after context compaction when the earlier system prompt may have been summarized away.

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5. Hook System

6 Hook Events

Event	Script	Matchers	Purpose
`PreToolUse`	`pretooluse.js`	9	Intercept and route tool calls before execution
`PostToolUse`	`posttooluse.js`	1 (all tools)	Capture session events (<20ms per call)
`PreCompact`	`precompact.js`	1	Build and store resume snapshot before compaction
`SessionStart`	`sessionstart.js`	1	Inject routing block + session guide on startup or compact resume
`UserPromptSubmit`	`userpromptsubmit.js`	1	Inject routing block on every prompt turn
`SubagentStop`	`subagent-stop.js`	1	Capture sub-agent outcomes into session DB

Total: 6 hook events, 14 matchers.

Cross-Platform Hook Execution

All hooks dispatch through `run-hook.cmd` — a polyglot CMD/bash wrapper:

- **Windows**: CMD portion calls `node hooks/<script>.js`
- **macOS/Linux**: Bash portion calls `node hooks/<script>.js`

The same file works on both platforms without any branch or platform check at the OS level.

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6. Main Skill & Decision Tree

The `context-mode` skill (`skills/context-mode/SKILL.md`) is the in-session authority for tool selection. It is auto-loaded by Cowork when context-mode is active.

Decision Tree

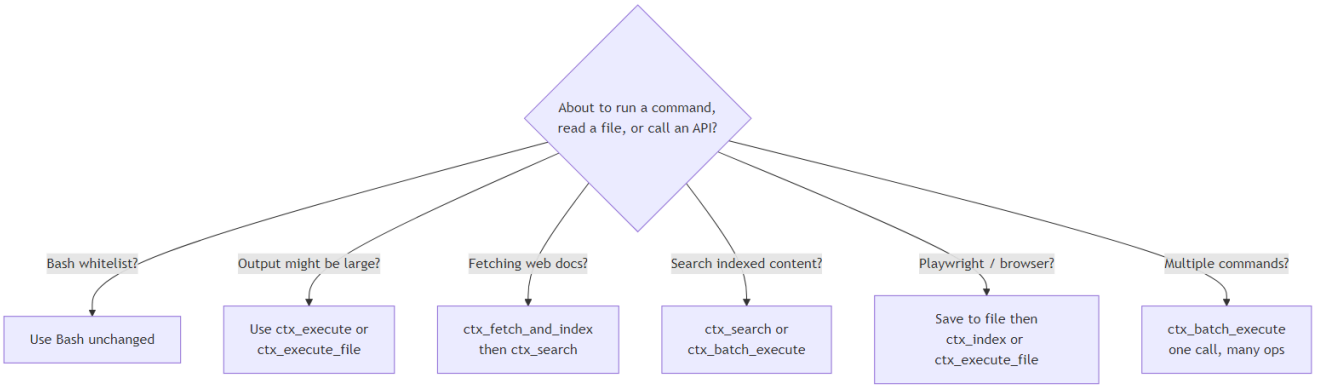

Figure 3: Decision Tree — tool selection based on operation type

Reference Files

The skill ships four reference files in `skills/context-mode/references/`:

- `anti-patterns.md` — what NOT to do (raw WebFetch, Read for analysis, LLM summarization calls)
- `patterns-javascript.md` — JS/TS patterns for common analysis tasks
- `patterns-python.md` — Python patterns for data processing
- `patterns-shell.md` — Shell patterns for system queries

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7. Bootstrapper (start.js)

The bootstrapper (`start.js`) is the MCP server entry point. It runs before `server/index.js` and handles:

Version Self-Healing

When Cowork caches a new version of the plugin, the bootstrapper detects it automatically:

1. Reads its own installation path from `__dirname`
2. Scans sibling version directories in the plugin cache
3. If a newer semver directory exists, updates `installed_plugins.json` to point to it
4. Next session starts on the newest version without user intervention

Dependency Self-Healing

On every startup:

1. `ensure-deps.js` checks that `better-sqlite3` native binaries match the current Node.js ABI
2. If a mismatch is detected (e.g., after a Node.js upgrade), it rebuilds the native module automatically
3. Pure-JS dependencies (`turndown`, `turndown-plugin-gfm`) are installed on demand if missing

This means the plugin survives Node.js upgrades and Cowork cache moves without user intervention.

Startup Sequence

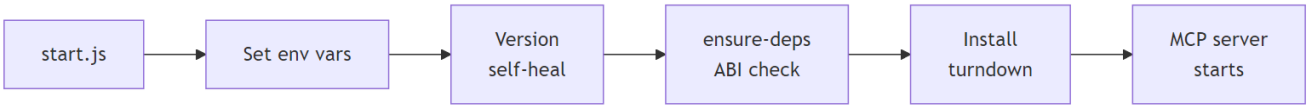

Figure 4: Startup Sequence — bootstrapper initialization chain

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8. CLAUDE.md System Directive

`CLAUDE.md` in the plugin root is loaded by Cowork into every session where context-mode is active. It functions as a standing system instruction, not a skill — it is always present regardless of what the user invokes.

What It Contains

Think in Code rule (mandatory): When Claude needs to analyze, count, filter, compare, search, parse, transform, or process data — it must write code via `ctx_execute` and `console.log()` only the answer. Raw data must never be read into context for mental processing.

Tool selection hierarchy:

1. `ctx_batch_execute` — primary gather tool (replaces many individual calls)
2. `ctx_search` — follow-up questions
3. `ctx_execute` / `ctx_execute_file` — processing and API calls
4. `ctx_fetch_and_index` + `ctx_search` — web documentation

Output constraints:

- Responses kept under 500 words
- Artifacts (code, configs) written to files — never returned as inline text
- Return only: file path + 1-line description

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9. Permission Rules (.claude/settings.json)

The `.claude/settings.json` file configures what Claude Code can and cannot do in the context-mode project itself. These rules apply when developing or testing the plugin.

```
{
  "permissions": {
    "deny": [
      "Bash(sudo *)",
      "Bash(rm -rf /*)",
      "Read(.env)",
      "Read(**/.env*)"
    ],
    "allow": [
      "Bash(git:*)",
      "Bash(ls:*)",
      "Bash(npm:*)",
      "Bash(npx:*)",
      "Bash(cat:*)",
      "Bash(echo:*)"
    ]
  }
}
```

Deny rules block dangerous or credential-exposing operations. Allow rules pre-approve common development commands so Claude doesn't prompt for permission on routine tasks.

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10. MCP Server & Tools

The MCP server (`server/index.js`) runs as a Node.js process communicating via stdio. It registers 9 tools:

Tool	Input	Output	Context Savings
`ctx_execute`	language, code, intent?	stdout only	94-100%
`ctx_execute_file`	files[], language, code	computed results	94-100%
`ctx_batch_execute`	commands[], queries[]	combined results + search	90-98%
`ctx_index`	content, source	chunk count confirmation	100% (content stored, not returned)
`ctx_search`	queries[], limit?	relevant snippets	N/A (retrieval)
`ctx_fetch_and_index`	url, queries?	cache hint or preview	95-99%
`ctx_stats`	(none)	session report	N/A
`ctx_doctor`	(none)	diagnostics	N/A
`ctx_purge`	confirm: true	confirmation	N/A

Auto-indexing Behavior

When `ctx_execute` output exceeds 5KB and an `intent` parameter is provided, the output is automatically indexed into the knowledge base and only the BM25-ranked snippets matching the intent are returned. When output exceeds 100KB without intent, it is indexed and a pointer is returned.

Progressive Search Throttling

`ctx_search` enforces a 60-second sliding window:

Calls in Window	Behavior
1-3	Full results (2 per query)
4-8	Reduced (1 per query) + warning
9+	Blocked, redirect to ctx_batch_execute

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11. Knowledge Base

Database Schema

```sql
-- Content storage
CREATE TABLE sources (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  label TEXT NOT NULL,
  chunk_count INTEGER,
  code_chunk_count INTEGER,
  indexed_at TEXT DEFAULT (datetime('now'))
);

-- Porter stemmer FTS5
CREATE VIRTUAL TABLE chunks USING fts5(
  title, content, source_id UNINDEXED, content_type UNINDEXED,
  tokenize='porter unicode61'
);

-- Trigram FTS5
CREATE VIRTUAL TABLE chunks_trigram USING fts5(
  title, content, source_id UNINDEXED, content_type UNINDEXED,
  tokenize='trigram'
);

-- Vocabulary for fuzzy correction
CREATE TABLE vocabulary (word TEXT PRIMARY KEY);
```

Chunking Strategy

- **Markdown**: Split on # headings, keep code blocks intact, max 4096 bytes per chunk, paragraph split fallback
- **JSON**: Recursive walk with key paths as titles, array items batched by size
- **Plain text**: Blank-line sections first, fallback to fixed-size line groups with 2-line overlap

TTL Cache

`ctx_fetch_and_index` checks `sources.indexed_at` before fetching. If age < 24 hours, returns a cache hint (~40 bytes) instead of re-fetching. Estimated bytes saved: `chunkCount * 1600`.

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12. Sandbox Executor

Process Isolation

- Each execution creates a unique temp directory: `mkdtempSync(join(OS_TMPDIR, '.ctx-mode-'))`
- Shell runs in project root (for git/relative paths); other languages run in temp dir

- Environment stripped of `NODE_OPTIONS` and `ELECTRON_RUN_AS_NODE`
- Hard stdout cap: 100MB (prevents `yes` or `/dev/urandom` from consuming memory)
- Default timeout: 30 seconds
- Background mode: detaches process after timeout, returns partial output

Platform-Specific Process Management

Platform	Process Kill Method
Windows	`taskkill /F /T /PID <pid>` (tree kill)
macOS/Linux	`process.kill(-pid, 'SIGKILL')` (process group)

Language-Specific Wrapping

Language	Wrapping
Go	Adds `package main` if missing
PHP	Adds `<?php` opening tag if missing
Shell	Adds `#!/usr/bin/env bash` + `set -e` if no shebang
Rust	Compile-then-run (rustc to binary, then execute)

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13. Session Continuity

Event Categories (13 types, 4 priority levels)

Priority	Category	Types
1 (Critical)	file	file_read, file_write, file_edit
1 (Critical)	task	task, task_update
1 (Critical)	rule	rule, rule_content
2 (High)	error	error_tool
2 (High)	cwd	cwd
2 (High)	env	env (package installs)
3 (Normal)	git	git (commit, push, merge, etc.)
3 (Normal)	subagent	subagent, subagent_complete
3 (Normal)	skill	skill
3 (Normal)	mcp	mcp_call
4 (Low)	data	data

Deduplication

SHA-256 hash of `type + category + data`, checked against last 5 events. Duplicates rejected.

FIFO Eviction

Max 1000 events per session. When exceeded, lowest-priority event is evicted first.

Compaction Snapshot

Budget: 2048 bytes. Sections included (priority order):

1. Files (P1) — last 10 active files with operation counts
2. Rules (P1) — CLAUDE.md and project rules
3. Task state (P1) — pending/in-progress tasks
4. Errors (P2) — last 5 errors
5. Git (P3) — recent git operations
6. Decisions (P3) — key decisions made
7. Subagents (P3) — sub-agent tasks
8. Environment (P4) — last cwd, env changes

Lower-priority sections are dropped first if budget is tight. Each section includes `ctx_search` hints for retrieving full details.

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14. Search Algorithm

Three-Layer Pipeline

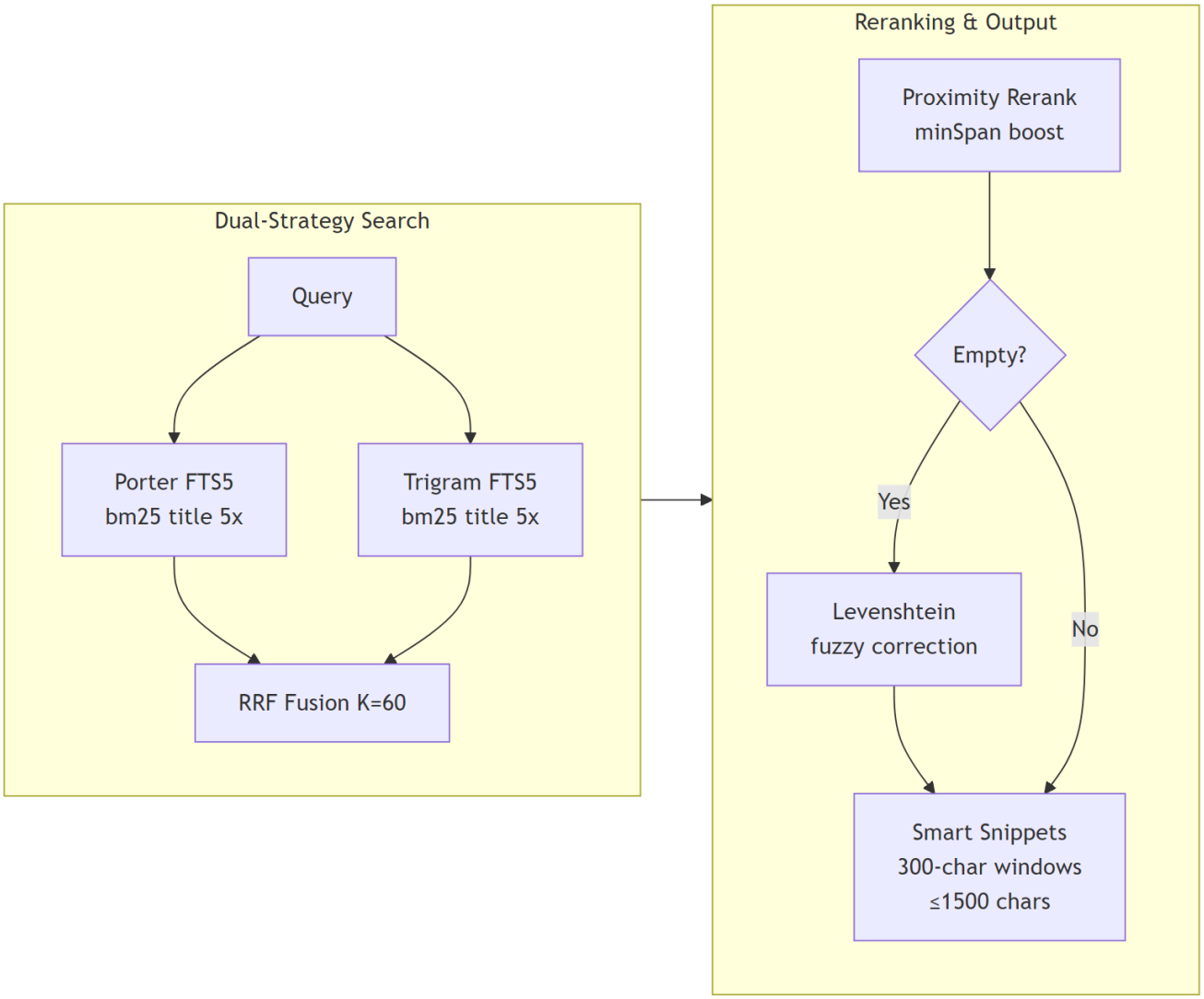

Figure 5: Search Algorithm — BM25 + trigram dual-strategy with RRF fusion

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15. Design Decisions

Why SQLite FTS5 over vector search?

FTS5 with BM25 provides deterministic, fast, and dependency-free full-text search. No embedding model needed, no API calls, no network dependency. The dual-tokenizer approach (Porter + trigram) with RRF fusion achieves search quality comparable to semantic search for code and technical documentation while running entirely locally.

Why subprocess isolation instead of in-process eval?

Security and resource control. Subprocess isolation prevents:

- Runaway code from blocking the MCP server
- Memory leaks from accumulating in the server process
- File system or network access from escaping the sandbox
- stdout/stderr from polluting the MCP transport

Why ≤2KB snapshot budget?

The snapshot is injected into context after compaction. Every byte of the snapshot is a byte that can't be used for the user's actual work. 2KB is enough to capture active files, pending tasks, recent errors, and search hints — the minimum Claude needs to resume effectively.

Why progressive search throttling?

Without throttling, Claude can fall into a search loop: search → not quite right → search again → refine → search again. Each search adds results to context. The throttle curve (2→1→blocked) forces batching, which is more context-efficient.

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16. Configuration Reference

Key Constants

Constant	Value	Purpose
MAX_CHUNK_BYTES	4096	Max bytes per FTS5 chunk
BM25 weights	bm25(chunks, 5.0, 1.0)	Title weighted 5x
RRF_K	60	Reciprocal Rank Fusion constant
TTL_MS	86,400,000 (24h)	Fetch cache TTL
SEARCH_WINDOW_MS	60,000 (60s)	Throttle window
SEARCH_BLOCK_AFTER	8	Block threshold
INTENT_SEARCH_THRESHOLD	5,000	Auto-index stdout threshold
HARD_CAP_BYTES	104,857,600 (100MB)	Executor stdout limit
MAX_EVENTS_PER_SESSION	1,000	FIFO eviction cap
DEDUP_WINDOW	5	Recent events checked for duplicates
MAX_SNAPSHOT_BYTES	2,048	Compaction snapshot budget

Environment Variables

Variable	Source	Purpose
`CLAUDE_PLUGIN_ROOT`	Cowork	Absolute path to plugin installation directory
`CLAUDE_PLUGIN_DATA`	Cowork	Persistent data directory (survives updates)
`CLAUDE_PROJECT_DIR`	Cowork	Current project working directory
`CLAUDE_SESSION_ID`	Cowork	Current session identifier
`NODE_PATH`	.mcp.json	Points to `${CLAUDE_PLUGIN_DATA}/node_modules`

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17. Platform Support

Windows

- Shell execution via Git Bash (`C:\Program Files\Git\usr\bin\bash.exe`)
- Process tree kill via `taskkill /F /T /PID`
- Temp directory: `%TEMP%` or `%TMP%`
- Runtime detection: `where` command (stderr suppressed)
- Hook execution: `run-hook.cmd` (CMD polyglot)

macOS

- Shell execution via `/bin/bash` or `/bin/zsh`
- Process group kill via `kill(-pid, SIGKILL)`
- Temp directory: `getconf DARWIN_USER_TEMP_DIR`
- Runtime detection: `command -v`
- Hook execution: `run-hook.cmd` (bash polyglot)

Linux

- Shell execution via `/bin/bash` or `/bin/sh`
- Process group kill via `kill(-pid, SIGKILL)`
- Temp directory: `$TMPDIR` or `/tmp`
- Runtime detection: `command -v`
- Hook execution: `run-hook.cmd` (bash polyglot)

Cross-Platform Guarantees

- All file paths use `path.join()`, never hardcoded separators
- Dynamic imports use `pathToFileURL()` for ESM compatibility on Windows
- better-sqlite3 ships pre-built binaries for both platforms via npm
- No platform-specific dependencies

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18. Testing

E2E Test Suite

The project ships a single comprehensive E2E test file (`test-e2e.js`) covering 207 tests across 18 sections:

Section	Coverage
1. Utils	sanitizeQuery, levenshtein, findMinSpan, extractSnippet, escapeXML
2. Exit Classify	Non-zero exit code interpretation (test runners, linters, signals)
3. Runtime Detection	Language runtime availability (node, python, shell, etc.)
4. Sandbox Executor	Subprocess isolation, stdout cap, timeout, language wrapping
5. Knowledge Base	FTS5 indexing, BM25 search, trigram search, RRF fusion, TTL cache
6. Session DB	Event storage, SHA-256 dedup, FIFO eviction, priority ordering
7. Snapshot Builder	Compaction snapshot within 2KB budget, section priority
8. Event Extraction	13 event categories, category detection accuracy
9. Routing Block	XML template content, command references
10. Hook .cmd Wrapper	Cross-platform execution
11. MCP Protocol Smoke Test	Live server via SDK client, all 9 tools respond
12. Plugin Discoverability	plugin.json, hooks.json, .mcp.json, directory structure
13. Spec Compliance	Claude Code plugin reference validation
14. OSS Attribution	Upstream credit in source headers and README
15. Plugin Manifest Validation	mcpServers field, flat .mcp.json format, required fields
16. PreToolUse Routing	curl→modify, git→passthrough, WebFetch→deny, Agent→modify
17. hooks.json Validation	6 hook events present, 9 PreToolUse matchers
18. Plugin CLAUDE.md and Settings	Directive content, permission rules structure

Running the Suite

```
cd context-mode
node test-e2e.js
```

The suite runs with no additional setup beyond `node install.js`. Output format: `PASS: <test name>` / `FAIL: <test name> - <detail>`. Exit code 0 = all pass, 1 = any failure.